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Rohin Patel

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Electric Vehicles – The Future?

Written by: Rohin Patel

Climate change targets, set by governments around the globe, have led to a widespread movement towards renewable energy, which is aimed at replacing the need for non-renewable sources of energy such as fossil fuels. One of the major contributors to the increased carbon emissions over the past few decades is the motoring industry as more and more cars on the road emit more carbon gasses. As such, the introduction of electric vehicles that are powered through renewable sources of energy is thought to be the solution to the climate change crisis. But are they as beneficial as sought out to be?

The introduction of electric vehicles (EVs) brought upon an era that shook up the automobile economy. But did it? Despite this general consensus, a recent analysis of American vehicle sales shows that gasoline cars were the key contributors with around 97 percent of sales being combustion vehicles (4).

Why Are People Not Transitioning To EVs?

While there are tax credits in more developed countries to help consumers finance the expense of purchasing an EV, the cost is still too high for the majority of the population. Most less-developed countries do not even have such tax credits to help finance the transition to EVs and, as such, only the rich and afford them in such areas. If EVs are priced similarly to equivalent traditional gas vehicles, more and more people would be likely to consider purchasing one (1).

Another major issue that prevents consumers from purchasing fully electric vehicles is the battery. The range of an EV is still considerably lower than that of an efficient gasoline-powered vehicle. Having a shorter range and a long charging time is a big disadvantage of owning an EV and drops the demand for them. In addition, the lifetime of an EV’s battery is still relatively unknown and to replace it with a new battery would set back an EV owner thousands of dollars. The majority of the population would not be willing to purchase a car that can potentially cause them thousands of dollars in replacement parts in a few years.

Moreover, one of the main challenges for all consumers of EVs is the lack of adequate infrastructures that support these vehicles such as charging stations. While there has been an increase in the number of charging stations in more developed nations, there is still a significant lack of such stations in South American, African and Asian countries (2).

A significant issue brought about by environmentalists against EVs is that even though they are produced to reduce carbon emissions, it uses electrical power that is generated using non-renewable sources of energy. Hence, it is thought to be a façade as it does not really lower the usage of fossil fuels and other non-renewable sources of energy.

Are EVs Any Good, Though?

The main argument that persuades millions of people to purchase EVs is the lack of need for gasoline. The average American spends upward of $3,000 on gas per year for each vehicle they own (3). Owning an EV completely cancels out that expense and while it may cost money to charge the EVs, it is significantly lower than what one would typically spend on a gasoline-powered vehicle. In addition, in more developed countries, there are plenty of tax benefits for owners of EVs that make these vehicles much more cost-effective.

Since electric vehicles tend to have a lower center of gravity due to large battery packs, they are more stable and safe on the road in the event of a collision. In case of an accident, the airbags in the EV will immediately deploy and the electric supply from the battery will be cut. This prevents serious injuries and fire, unlike most gasoline cars (5).

The convenience of an EV is often regarded as a major selling point for EV manufacturers such as Tesla. Unlike gas stations for traditional gasoline-powered vehicles, an EV does not need to go to a specific location to recharge before use. It can be conveniently be charged at one's home through a normal household electric socket.

How Do We Get More On The Road?

It is clear that the huge potential of these EVs and their massive help towards a more green and environmentally-friendly world is yet to be properly manifested. How can we make this happen? Firstly, it is vital to get less developed countries to create incentives for their population to purchase EVs and go green. The majority of the world's population lives in less developed countries where there are no incentives to purchase EVs and very minimal infrastructure to help support consumers of EVs. If EV manufacturers lobby such proposals to governments of less developed countries they are likely to get a greater demand for electric vehicles in these countries. Moreover, even though there have been efforts to improve infrastructure to support EVs in developed nations, they are still significantly fewer charging points than gas stations and until that disparity evens out it will hinder consumers from purchasing EVs.

Also, the use of non-renewable energy for the generation of electricity is a major factor that deters environmentally conscious consumers and, as such, if governments can commit to new methods of electricity generation such as wind or solar power, they are likely to attract millions of consumers towards more green transportation modes. As such, to meet climate change targets through increased EVs on the road, governments are required to take on initiatives that will allow for all of the previous issues to be solved. For now, gasoline-powered vehicles still account for the lion's share of the roads.

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